The empirical recurrence for OEIS A268892: recovering a conjecture that is not written down

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Abstract

OEIS A268892 records "Empirical recurrence of order 68", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 177$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 68, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 68 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A268892 is "Number of 7Xn binary arrays with some element plus some horizontally or antidiagonally adjacent neighbor totalling two exactly once.". It has offset 1 and begins

$$0, 2285, 60666, 1706732, 39670270, 911930096, 19876401224, 426591159751, \ldots$$

The entry states:

> Empirical recurrence of order 68 (see link above)

As of the "Last modified" line on the live entry (Feb 15 2016 11:38:51, revision 4) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 256$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 177$ of the 256, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 177$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 354$ exact terms of the model returns order 68 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{68} c_i\, a(n-i),$$

c_1	2	c_{18}	173999731178900	c_{35}	-342892193937831224	c_{52}	6831947795820
c_2	575	c_{19}	752894468160068	c_{36}	376076619803764875	c_{53}	-3691011187128
c_3	2258	c_{20}	-1619456519331548	c_{37}	122344533435654994	c_{54}	-667732708590
c_4	-108176	c_{21}	-2974851765609104	c_{38}	-266948400066666565	c_{55}	277738545544
c_5	-911990	c_{22}	9965821486840556	c_{39}	-1415361918802462	c_{56}	53273491922
c_6	5033607	c_{23}	5297939562752408	c_{40}	131286747352704128	c_{57}	-14532193700
c_7	67202314	c_{24}	-40773860905086755	c_{41}	-24611270856848094	c_{58}	-3237657340
c_8	-102518581	c_{25}	11819016195409426	c_{42}	-47048921337207887	c_{59}	477091580
c_9	-2546860380	c_{26}	108099916599164729	c_{43}	15363171956845622	c_{60}	137735787
c_{10}	1071160792	c_{27}	-102292737919588474	c_{44}	12678090389802021	c_{61}	-7183986
c_{11}	62403960868	c_{28}	-170897153299813112	c_{45}	-5562417498600320	c_{62}	-3691209
c_{12}	-13978070470	c_{29}	301311104540569846	c_{46}	-2637637855752328	c_{63}	-59946
c_{13}	-1077916344448	c_{30}	113148118427075239	c_{47}	1400242908170224	c_{64}	52520
c_{14}	483856639070	c_{31}	-505321745373284666	c_{48}	436186414185796	c_{65}	3574
c_{15}	13475140095656	c_{32}	105968159348257723	c_{49}	-258896733279652	c_{66}	-245
c_{16}	-11903638787628	c_{33}	527126204078587584	c_{50}	-59321335771432	c_{67}	-34
c_{17}	-120979588220176	c_{34}	-327853166225767520	c_{51}	35795027808888	c_{68}	-1

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 68, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $68+4$ terms removed returns order 68 as well, so $\deg q_{\min} = 68 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $68+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 68 that this sequence satisfies — and that family turns out to have exactly one member.

References

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